

Probabilistic Health-Risk Assessment of Arsenic in Bangladesh Groundwater Using Monte Carlo Simulation

Section B • Group 3

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Abstract

Estimating the health effects of arsenic-contaminated groundwater is difficult because arsenic levels and human exposure vary widely. We adapted the Monte Carlo framework of Yadav and Kalkal to Bangladesh using 810 groundwater samples from ten districts, together with survey-weighted body-weight records for 8,013 adults and 22,576 children. Rather than replacing measurements below the detection limit with fixed values, we treated them as left-censored data and constructed reverse Kaplan–Meier empirical distributions. We then fitted district-specific Normal, Lognormal, and Gamma models. Adult body weight was represented by a Lognormal distribution, while child body weight followed a lower-truncated Normal distribution. For each district and population group, we propagated 10,000 inverse-CDF samples through ingestion and dermal-exposure equations to estimate the hazard index (HI) and excess lifetime cancer risk (ELCR). Comilla had the highest estimated risk: its P95 HI was 51.3 for adults and 259.6 for children, and the corresponding probabilities of $HI > 1$ were 72.3% and 85.1%. Children had a higher HI in every district. Arsenic concentration was the dominant source of risk variation ($\rho_s = 0.950\text{--}0.998$). Although 10,000 iterations were sufficient for all $P(HI > 1)$ estimates, bootstrap analysis showed that uncertainty in the fitted input distributions was much larger than Monte Carlo sampling error. These findings support targeted arsenic mitigation and continued district-level monitoring.

Keywords

arsenic, Bangladesh, groundwater, health risk assessment, Monte Carlo simulation, censored data, uncertainty analysis

1 Introduction and Base Paper Review

Long-term exposure to inorganic arsenic in drinking water can lead to serious non-cancer and cancer outcomes. The problem is especially important in Bangladesh, where groundwater is widely used for drinking and arsenic levels can change sharply over short distances. The DPHE/BGS National Hydrochemical Survey measured 3,534 boreholes across 61 districts [2]. A conventional point estimate, however, cannot capture this spatial variation, measurements below detection limits, or differences in exposure across the population.

Our base paper, by Yadav and Kalkal, compared deterministic and probabilistic arsenic-risk assessments for seven districts

Table 1: Analysis datasets and retained records.

Input	Processing summary	Retained
Arsenic	Ten districts; detected and left-censored values; $\mu\text{g L}^{-1}$ converted to mg L^{-1}	810
Adult BW	STEPS ages 18–69; valid measured weight and survey weight	8,013
Child BW	MICS ages 0–59 months; special codes and WAZ-flagged weights removed	22,576

in Punjab, India [1]. The authors propagated distributions for arsenic concentration, water ingestion, exposure frequency and duration, body weight, and exposed skin area through U.S. EPA ingestion and dermal-exposure equations. They reported the non-carcinogenic hazard index (HI), excess lifetime cancer risk (ELCR), 95th-percentile risk, and Spearman sensitivity coefficients. Their analysis found $HI > 1$ for children in every district and identified arsenic concentration as the most influential input.

We build on that framework by replacing the Punjab arsenic and body-weight inputs with Bangladesh data. We also treat censoring, survey weights, convergence, and uncertainty more explicitly. Where a suitable Bangladesh-specific exposure distribution was unavailable, we retained the assumption used in the base paper. Our final analysis covers ten districts and separately evaluates adults and children. The code and supporting materials needed to reproduce the analysis are available at https://github.com/fa88923/Bangladesh_Arsenic_Health_Risk_Assessment.

2 Methodology

2.1 Data Sources and Preprocessing

We combined three data sources (Table 1). Groundwater arsenic measurements came from the DPHE/BGS/DFID National Hydrochemical Survey, released in 2000 and documented by BGS [2]. Adult body weights came from the WHO Bangladesh STEPS 2018 public-use dataset [3], and child body weights came from Bangladesh MICS7 2025, conducted by the Bangladesh Bureau of Statistics with UNICEF [4]. We kept the raw files unchanged and recorded a SHA-256 hash for each file.

Values reported as strings such as “< 6” were treated as left-censored observations. We did not replace them with zero, the detection limit, or half the detection limit. Across the ten districts, the data contained 524 detected values and 286 censored values.

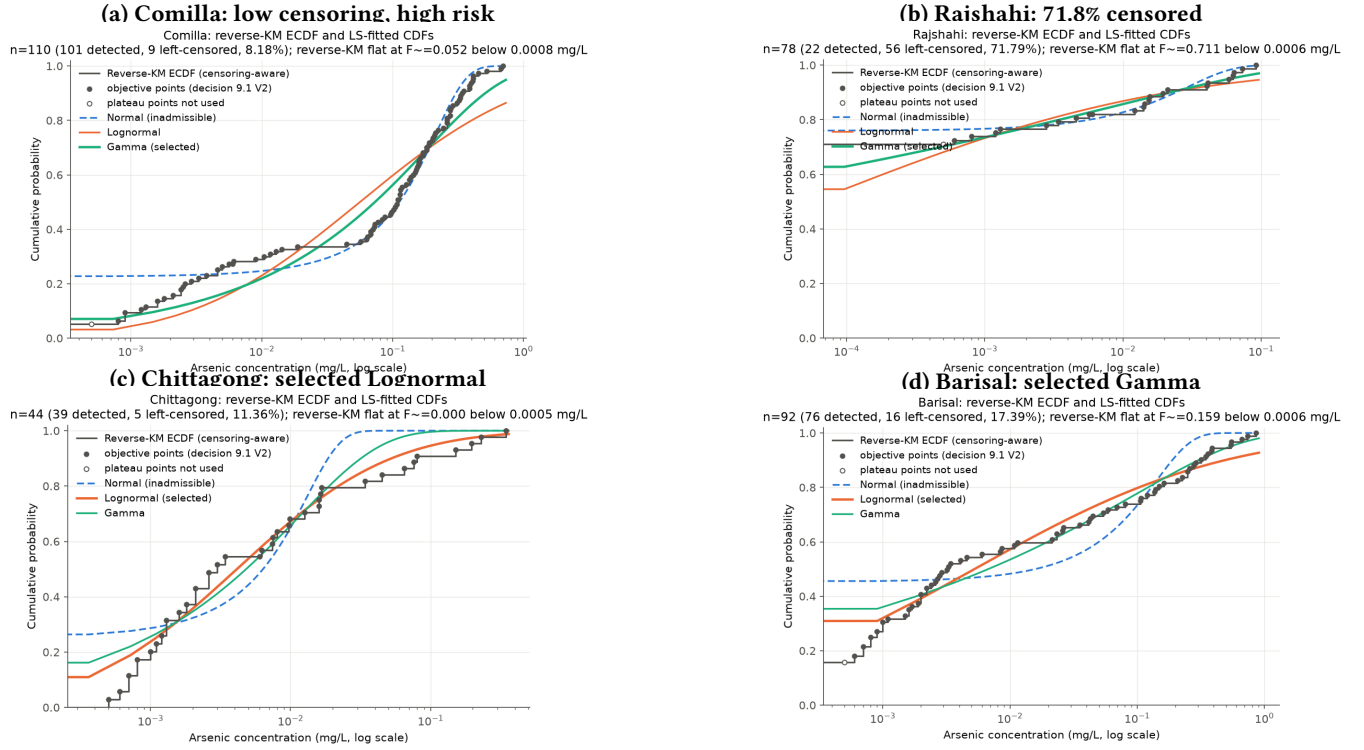


Figure 1: Representative censoring-aware empirical arsenic CDFs and fitted candidate distributions. Full diagnostics were generated for all ten districts.

Because 71.8% of the Rajshahi observations were censored, we flag its results as particularly uncertain. We kept repeated coordinates when the sampling date or well depth differed, and used the provided survey weights in both body-weight datasets without trimming valid tail observations.

2.2 Distribution Fitting

For each district, we used the reverse Kaplan–Meier estimator to construct an empirical arsenic CDF that accounts for censoring. We fitted Normal, Lognormal, and Gamma candidates to this CDF by least squares, then evaluated them using censored-likelihood AIC, bootstrap stability, quantile agreement, and upper-tail behavior. Gamma was selected for eight districts, while Lognormal was selected for Chittagong and Bogra (Table 2). We initially considered a Lognormal model for Barisal, but rejected it because its fitted P99 was 26 times the empirical P99 and 7.5% of simulated values exceeded the largest observed concentration.

For body weight, we combined tied observations and converted the normalized survey masses q_j into weighted midpoint plotting positions:

$$p_j = \sum_{r < j} q_r + \frac{q_j}{2}. \quad (1)$$

We estimated each candidate model by minimizing

$$\sum_j q_j [F(x_j; \theta) - p_j]^2. \quad (2)$$

Table 2: Final stochastic models. Parameters for Gamma are shape k and scale θ ; Lognormal parameters are $\mu_{\log}$ and $\sigma_{\log}$.

Input	Family	District/population
Arsenic	Gamma	Dhaka, Rajshahi, Khulna, Barisal
		Sylhet, Rangpur, Mymensingh, Comilla
Arsenic	Lognormal	Chittagong, Bogra
BW	Lognormal	Adult
BW	Truncated Normal	Child

The selected adult model was

$$BW \sim \text{Lognormal}(\mu_{\log} = 4.0024, \sigma_{\log} = 0.2002).$$

For children, a Normal parent distribution with $\mu = 10.8570$ and $\sigma = 3.2281$ was truncated below 1.6 kg, the smallest retained measurement. This choice reproduced the dose-relevant $E(1/BW)$ within 0.1% and reduced the worst P1–P99 quantile error from 36% under the provisional Gamma model to 1.7%.

Figure 1 compares representative empirical and fitted arsenic CDFs. The clear differences among districts support the use of separate exposure models instead of a single pooled national distribution.

2.3 Risk Equations

We implemented the base paper’s equations directly, without algebraic simplification. Let C denote arsenic concentration; IR , ingestion rate; EF , exposure frequency; ED , exposure duration; BW , body weight; AT , averaging time; K_p , skin permeability; ET , exposure time; SA , exposed skin area; and CF , the unit-conversion factor. The resulting equations are

$$ADD_{ing} = \frac{C \cdot IR \cdot EF \cdot ED}{BW \cdot AT}, \quad (3)$$

$$ADD_{dermal} = \frac{C \cdot K_p \cdot EF \cdot ED \cdot ET \cdot SA \cdot CF}{BW \cdot AT}, \quad (4)$$

$$HQ_{ing} = \frac{ADD_{ing}}{RfD_{ing}}, \quad HQ_{dermal} = \frac{ADD_{dermal}}{RfD_{dermal}}, \quad (5)$$

$$HI = HQ_{ing} + HQ_{dermal}, \quad (6)$$

$$ELCR = ADD_{ing,cancer} \cdot CSF. \quad (7)$$

The reference doses were

$$RfD_{ing} = 3.0 \times 10^{-4}, \quad RfD_{dermal} = 2.85 \times 10^{-4} \quad (8)$$

in $\text{mg kg}^{-1} \text{d}^{-1}$. We also used $CSF = 1.5 (\text{mg kg}^{-1} \text{d}^{-1})^{-1}$, $K_p = 0.001 \text{ cm h}^{-1}$, and $CF = 10 \text{ L cm}^{-1} \text{ m}^{-2}$. For non-cancer risk, we set $AT = ED \times 365$. For the primary ELCR estimate, we used the conventional 70-year lifetime average, $AT_{cancer} = 25,550$ days [5]. We also retained the base paper’s $ED \times 365$ convention as a comparison scenario.

We interpreted the base paper’s “mean $\pm$ SD” Lognormal inputs as arithmetic moments and converted them to log-space. Treating the reported values as log-space parameters directly would generate physically implausible adult skin areas and would not reproduce the paper’s results. Exposure frequency followed Triangular(180, 345, 365) days/year. Exposure time followed Triangular(0.13, 0.20, 0.33) hours/day, and child skin area followed Triangular(0.29, 0.60, 0.95) m^2 .

2.4 Monte Carlo, Convergence, and Sensitivity

For each district–population pair, we generated six independent uniform random variables and transformed them through the relevant inverse CDFs to sample C , IR , BW , EF , ET , and SA . The primary analysis used $N = 10,000$ with master seed 20260923. We retained all iteration-level values for the 20 primary runs and summarized the mean, median, P95, $P(HI > 1)$, $P(HI > 2)$, and $P(ELCR > 10^{-4})$.

To assess convergence, we tested $N \in \{1,000, 5,000, 10,000, 20,000\}$. We used five independent random streams per size, producing 400 runs and 3.6 million iterations in total. We considered a continuous statistic stable when both its change between successive sample sizes and its between-seed coefficient of variation were no greater than 5%. For probabilities, both differences had to remain within one percentage point.

We measured sensitivity with Spearman rank correlations calculated from the primary simulation draws. To separate fitted-parameter uncertainty from normal input variability, we generated 1,000 bootstrap fits for arsenic and 1,000 Rao–Wu survey bootstrap fits for each body-weight population. Common random numbers helped isolate the effect of changing model parameters. We also tested six model scenarios covering distribution-family selection,

the under-5 body-weight age gap, and alternative interpretations of the base paper’s Lognormal parameters.

3 Implementation and Experiments

We implemented the workflow in Python with NumPy, pandas, SciPy, and Matplotlib, and stored tabular outputs in Parquet format [6–9]. Vectorized array operations allowed us to run the simulations efficiently. Each run received a deterministic random-number stream derived from the master seed together with the district, population, sample size, and replicate number. Run manifests recorded input hashes, model parameters, software versions, seeds, output hashes, and row counts.

We validated the workflow by checking data reconciliation, unit conversions, physical bounds, equation identities, factor-by-factor scaling, inverse-CDF sampling, and exact same-seed reproduction. Our implementation reproduced 25 of the 28 values in the base paper’s deterministic Table 4 to within 0.5%. The remaining three ELCR values differ from calculations based on the paper’s own printed inputs by 1.2–6.7%, suggesting typographical errors in the published table. At the WHO guideline concentration of 0.01 mg L^{-1} , our deterministic Bangladesh benchmark gave $HI = 1.20$ for adults and $HI = 5.53$ for children. Both values are already above the non-cancer threshold of one.

4 Results and Discussion

4.1 Primary Risk Estimates

Table 3 summarizes the primary P95 HI, lifetime-averaged P95 ELCR, and $P(HI > 1)$. Comilla had the highest estimated risk for both adults and children, followed by Barisal, Dhaka, and Khulna. Rangpur and Rajshahi had the lowest P95 HI values, although Rajshahi’s 71.8% censoring means that its fitted distribution should be interpreted cautiously.

Table 3: Primary Monte Carlo results at $N = 10,000$. Entries are adult/child; ELCR is scaled by 10^{-3} .

District	P95 HI	P95 ELCR	$P(HI > 1)$ (%)
Dhaka	21.6/121.1	9.68/4.67	33.5/47.7
Chittagong	6.66/34.7	2.99/1.34	23.2/53.8
Rajshahi [†]	3.17/18.2	1.42/0.699	11.2/20.3
Khulna	15.7/84.3	7.03/3.25	30.0/44.9
Barisal	32.1/175.1	14.4/6.75	41.2/55.8
Sylhet	6.61/35.8	2.96/1.38	32.8/60.0
Rangpur	1.84/9.62	0.827/0.371	10.0/29.9
Mymensingh	7.19/37.9	3.23/1.46	19.1/32.0
Comilla	51.3/259.6	23.0/10.0	72.3/85.1
Bogra	6.57/33.3	2.95/1.28	15.5/32.4

ELCR uses a 70-year lifetime averaging time. [†] 71.8% of Rajshahi observations were left-censored.

In every district, the child P95 HI was 5.1–5.7 times the adult value. The main reason is that the children in the model weighed about one-fifth as much as the adults while their assumed water intake remained comparable. The lifetime-averaged child ELCR was nevertheless lower because six years of exposure were averaged over a 70-year lifetime. If we instead apply the base paper’s $AT = ED \times 365$ convention, child ELCR becomes 11.7 times the primary child estimate and exceeds adult ELCR by approximately the same factor as HI. This difference in averaging time is important when comparing cancer-risk results across studies.

Ingestion accounted for almost all of the estimated risk; dermal exposure contributed only 0.1–0.3% of mean HI. Because the arsenic distributions were strongly right-skewed, the probabilistic P95 values were 1.8–5.4 times the corresponding deterministic estimates. At the same time, the median was below the deterministic estimate in every district. A calculation based only on mean inputs therefore represents neither a typical exposure nor a plausible upper-tail exposure.

Simulated HI distribution, Comilla (primary run)

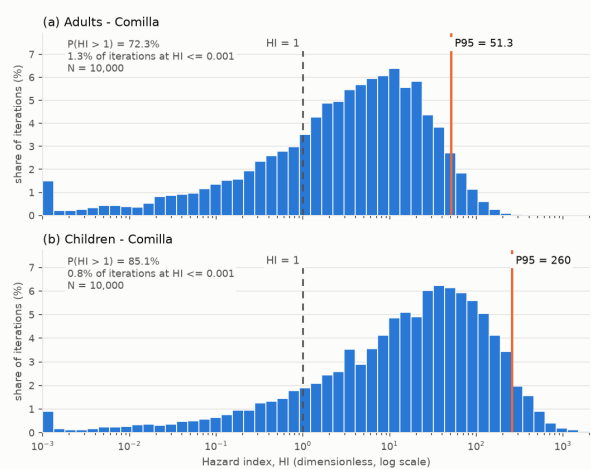


Figure 2: Representative adult and child HI distributions for high-risk Comilla (top) and lower-risk Rangpur (bottom). The dashed line marks $HI = 1$ and the orange line marks P95.

4.2 Sensitivity and Uncertainty

Arsenic concentration dominated the sensitivity results for all 20 district–population combinations, with $\rho_s = 0.950$ – 0.998 . Ingestion rate was the second most influential input (0.054–0.231), while body weight had the expected negative relationship with risk (–0.022

Table 4: P95 HI with 95% fitted-parameter intervals.

District	Adult	Child
Dhaka	21.6 [10.2, 31.2]	121 [58, 174]
Chittagong	6.7 [1.6, 24.8]	35 [9, 123]
Rajshahi	3.2 [1.1, 6.2]	18 [7, 35]
Khulna	15.7 [7.5, 24.6]	84 [40, 133]
Barisal	32.1 [9.6, 51.8]	175 [52, 284]
Sylhet	6.6 [4.1, 9.7]	36 [22, 52]
Rangpur	1.8 [1.0, 3.0]	9.6 [5.0, 15.6]
Mymensingh	7.2 [2.7, 11.0]	38 [14, 59]
Comilla	51.3 [40.6, 65.1]	260 [212, 330]
Bogra	6.6 [0.8, 13.8]	33 [4, 70]

to –0.157). Exposure frequency had only a small effect. Exposure time and skin area had correlations that were statistically indistinguishable from zero because the dermal pathway contributed less than 0.3% of HI. The base paper reported lower arsenic correlations, approximately 0.65–0.90. The stronger relationship in our study is consistent with the wider arsenic distributions observed across the selected Bangladesh districts.

Uncertainty in the fitted parameters was much greater than numerical Monte Carlo error (Table 4). For example, the adult P95 HI for Chittagong was 6.7, but its 95% bootstrap interval ranged from 1.6 to 24.8. Comilla had the narrowest relative uncertainty because it had the largest district sample and only 8.2% censoring. The adult and child intervals had similar relative widths, indicating that the main limitation was the 44–110 arsenic observations available per district, rather than the much larger body-weight samples.

Across the alternative model scenarios, $P(HI > 1)$ was more stable than either P95 or the mean. Changing the arsenic distribution family altered P95 by as much as a factor of 3.14 and the mean by as much as a factor of 46, yet changed $P(HI > 1)$ by no more than 19%. Extending the child body-weight distribution to cover ages 0–71 months lowered child P95 by 5.5–8%, suggesting that the under-5 dataset introduces a small conservative bias. Replacing the truncated Normal child model with a Gamma distribution changed P95 by only 1.5–5%. For this reason, threshold exceedance is our most defensible headline measure; P95 and mean estimates should always be reported with an explicit warning about upper-tail model choice.

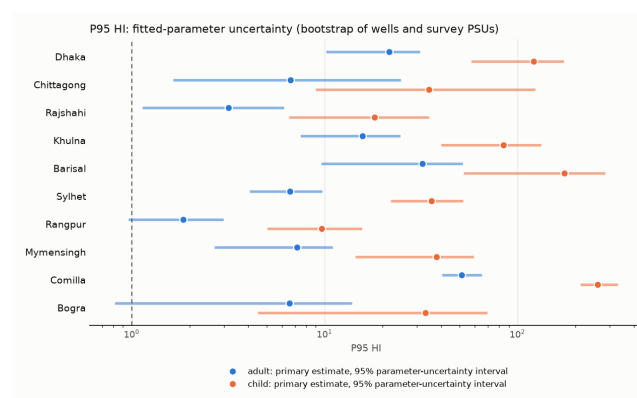


Figure 3: Fitted-parameter uncertainty for adult and child P95 HI.

4.3 Convergence and Numerical Reliability

With $N = 10,000$, $P(HI > 1)$ satisfied the convergence criterion in all 20 district–population combinations, and its Monte Carlo standard error never exceeded 0.005. P95 HI and P95 ELCR met the criterion in 16 combinations. Bogra adults, Bogra children, Chittagong children, and Sylhet adults required $N = 20,000$. Even for these cases, the difference between the 10,000- and 20,000-iteration P95 estimates was no greater than 9.1%. Mean HI for Bogra did not converge within the tested grid because the selected Lognormal arsenic model has a very heavy upper tail; its median and P95 were considerably more stable. Overall, the numerical error decreased approximately as $N^{-1/2}$ (Figure 4).

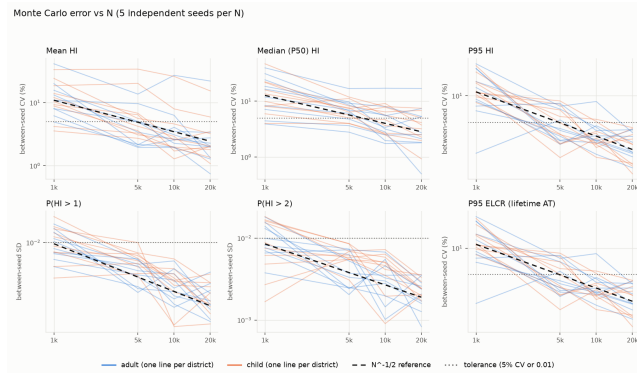


Figure 4: Between-seed Monte Carlo error across the convergence grid.

4.4 Comparison with the Base Paper

Our findings agree with the base paper on two central points: children face greater non-cancer risk, and arsenic concentration drives most of the variation in risk. Our analysis extends the original study by treating non-detects as censored observations, fitting a separate arsenic distribution for each district, deriving body-weight models from weighted national surveys, and separating input variability from numerical, parameter, and model-choice uncertainty. We also distinguish a conventional lifetime-averaged ELCR from the base paper’s exposure-duration averaging approach. These additions reinforce the original qualitative conclusion, while showing that the size of upper-tail risk estimates remains sensitive to the available data and the chosen model.

5 Limitations

This study has several limitations. First, we treated the six sampled inputs as independent because no dataset links individual people to specific wells. The BGS measurements describe the sampled wells, not a population-weighted census of exposure. Values below the smallest detection limits also depend on distributional extrapolation. This choice mainly affects medians and lower percentiles rather than threshold exceedance. Data limitations are especially important for Rajshahi, which has only 22 detected observations. The upper tail of Bogra’s Lognormal model is unstable, Rangpur’s Gamma model may underestimate the most extreme values, and the selected Chittagong model underestimates the empirical P95.

Second, the distributions for water intake, exposure frequency, exposure time, and skin area came from the Punjab study or EPA guidance rather than direct measurements from Bangladesh. The child body-weight dataset covers ages 0–59 months, whereas the assumed child exposure duration is six years. Finally, we modeled total arsenic only; the analysis does not distinguish arsenic species or account for other groundwater contaminants.

6 Conclusion and Future Work

Our Bangladesh-focused analysis found substantial arsenic-related health risk in all ten districts. Comilla consistently had the highest estimates, and children faced much greater non-cancer risk than adults. Arsenic concentration explained nearly all of the rank variation in HI, while ingestion contributed far more risk than dermal exposure. The main $P(HI > 1)$ estimates were numerically stable with 10,000 iterations. However, the bootstrap results show that limited well samples and assumptions about distribution tails contribute far more uncertainty than the Monte Carlo calculation itself. These findings support practical measures such as arsenic removal, identification of safe wells, regular district-level monitoring, and additional protection for children.

Future work should collect Bangladesh-specific information on water intake and daily activities, connect households to the wells they actually use, and model spatial dependence and correlations among inputs. Body-weight data for children aged 60–71 months would remove the current age-coverage gap. Future analyses could also include arsenic species and other groundwater contaminants. Hierarchical or spatial models for censored data may help share information across districts with small samples while preserving local differences. Most importantly, additional groundwater measurements would reduce uncertainty more effectively than simply increasing the number of Monte Carlo iterations; the current analysis is limited by data, not computing power.

Code and Reproducibility

The complete source code, configuration files, documentation, and instructions for reproducing the analysis are available at https://github.com/fa88923/Bangladesh_Arsenic_Health_Risk_Assessment.

References

- [1] Sangeeta Yadav and Sunil Kalkal. 2024. Health risk assessment using Monte-Carlo simulations due to arsenic contamination in groundwater in Punjab. *Journal of Water and Health* 22, 12 (2024), 2304–2319. <https://doi.org/10.2166/wh.2024.188>
- [2] British Geological Survey and Department of Public Health Engineering. 2001. *Arsenic Contamination of Groundwater in Bangladesh: DPHE/BGS National Hydrochemical Survey*. British Geological Survey, Keyworth, UK. <https://www.bgs.ac.uk/groundwater/quality/groundwater-and-health/arsenic-contamination-of-groundwater/bangladesh-reports/>
- [3] National Institute of Preventive and Social Medicine and World Health Organization. 2018. *Bangladesh STEPS Survey 2018: Public-Use Dataset and Country Report*. <https://extranet.who.int/ncdsmicrodata/index.php/catalog/770>
- [4] Bangladesh Bureau of Statistics and UNICEF Bangladesh. 2026. *Progoti Pathay: Bangladesh Multiple Indicator Cluster Survey 2025, Survey Findings Report*. BBS and UNICEF Bangladesh, Dhaka, Bangladesh.
- [5] United States Environmental Protection Agency. 1989. *Risk Assessment Guidance for Superfund, Volume I: Human Health Evaluation Manual (Part A)*. EPA/540/1-89/002, Washington, DC.
- [6] Charles R. Harris et al. 2020. Array programming with NumPy. *Nature* 585 (2020), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- [7] Pauli Virtanen et al. 2020. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17 (2020), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>

- [8] Wes McKinney. 2010. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, 56–61.
- [9] John D. Hunter. 2007. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering* 9, 3 (2007), 90–95. <https://doi.org/10.1109/MCSE.2007.55>